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Quote Competition in Corporate Bonds

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1 Big picture

- **Research question.** The paper asks whether indicative dealer quotes, or “runs,” matter in an over-the-counter corporate bond market in which quotes are nonbinding, pre-trade transparency is limited, and trading is often relationship based.
- **Setting.** The authors use BondCliQ quote data matched to regulatory TRACE trades for U.S. corporate bonds from October 2019 to May 2020. The sample contains 8.5 million bid and ask quotes from 35 dealers on roughly 24,000 bonds.
- **Core finding.** Dealer quotes, despite being indicative rather than binding, lower client trading costs and increase both dealer-level and bond-level trading volume.
- **Dealer behavior.** Dealers quote to advertise trading intent, manage inventories, and attract order flow. Lead underwriters and dealers with inventory motives are more likely to quote.
- **Client behavior.** Institutional investors use quotes to trade with better-priced dealers, including dealers outside their established relationships.
- **Imperfect competition.** Quote competition is not perfect: the best quote often fails to win the trade, and trade-throughs are common.
- **Negotiation channel.** Quotes affect bargaining. Dealers who posted worse quotes can still improve prices in bilateral negotiations, especially when clients can use competing quotes as outside options.
- **Aggregate market effect.** Quoting is not purely zero-sum redistribution across dealers. More bond-level quote supply is associated with more bond-level trading and better bond-level execution quality.
- **Relationship channel.** Quotes partly substitute for client-dealer relationships. Nonrelationship dealers are more likely to win insurance-company order flow when they quote, and quote quality further increases this switching.
- **Takeaway.** Indicative quotes can meaningfully improve OTC market functioning even without a centralized limit order book or binding displayed quotes.

2 Incremental contribution of the paper

2.1 Contribution relative to search-and-matching theories of OTC markets

- Prior theories emphasize search frictions, limited counterparty discovery, and relationship-based bargaining in OTC markets.
- This paper directly observes a pre-trade information object, dealer runs, that can reduce those frictions.
- It shows that quotes affect who trades, how much trading occurs, and at what prices trades execute.

Interpretation: The incremental contribution is to move from an inferred-search-friction view of OTC markets to direct evidence on the informational role of pre-trade quotes.

2.2 Contribution relative to empirical corporate bond trading studies

- Earlier work often studies post-trade TRACE data and infers liquidity from transaction prices, spreads, and trading frequency.
- The paper links dealer-level quotes to dealer-level trades, which allows a much sharper test of whether quotes guide order flow and execution quality.
- The matching of BondCliQ quotes to TRACE trades gives the authors a micro-level view of the quote-trade chain.

Interpretation: The paper’s data contribution is substantial: it observes the pre-trade market information that investors and dealers see before negotiating trades.

2.3 Contribution relative to relationship-banking and relationship-trading literature

- Relationship-based trading can lower search costs but may also reduce competition and keep clients captive.
- The paper shows that quotes allow nonrelationship dealers to compete for order flow.
- In insurance-company trades, nonrelationship dealers handle 23% of trades when they quote, compared with 11% when they do not quote.

Interpretation: The paper adds a mechanism through which transparency can weaken relationship lock-in without eliminating relationships entirely.

2.4 Contribution relative to price transparency and market design

- In centralized markets, public binding quotes create a clear best-price benchmark.
- In this OTC setting, quotes are indicative, privately distributed, and nonbinding.
- The paper shows that even nonbinding quotes can discipline bargaining, but not fully: trade-throughs remain frequent.

Interpretation: The paper’s incremental insight is that partial transparency improves outcomes, but the welfare effects are limited by quote leakage, incomplete dissemination, and ex post bargaining.

3 Data and measurement

3.1 BondCliQ quote data

- BondCliQ collects dealer runs by joining dealers’ distribution lists and redistributes the quotes to subscribers.
- A quote includes a dealer name, timestamp, CUSIP, side of the market, and quoted bid or ask price or yield spread.
- The authors identify dealer quote messages and construct dealer-bond-day quote measures.
- The main sample contains roughly 8.5 million quotes from 35 dealers between October 2019 and May 2020.

Interpretation: The quote data capture the pre-trade information environment of institutional corporate bond trading. This is unusual because dealer runs are proprietary and normally unobservable.

3.2 TRACE trade data

- Trades come from supervisory TRACE, which contains dealer identities.
- The authors match TRACE trades to BondCliQ quotes by dealer and bond.
- Trades by CliQ and non-CliQ dealers are both used for aggregate bond-level analyses.

Interpretation: The dealer identifier is crucial. It allows the paper to ask whether the dealer posting a quote later trades the same bond and whether the quote affects the execution price.

3.3 Insurance-company trades and relationship data

- To study relationship substitution, the paper uses insurance-company transaction data matched to dealer identities.
- Relationship dealers are defined using the insurer’s historical counterparty distribution before the sample period.
- Dealers in the top half of a given insurer’s counterparties are classified as relationship dealers; other dealers are nonrelationship dealers.

Interpretation: This data set allows a direct test of whether quotes draw clients away from established dealer relationships.

3.4 Core quote and trade variables

- $HasQuote_{d,b,t}$ equals one if dealer d quotes bond b on day t .
- $HasTrade_{d,b,t}$ equals one if dealer d trades bond b on day t .
- $LogNumQuotes_{d,b,t} = \log(1 + NumQuotes_{d,b,t})$ measures quote intensity.
- $LogNumTrades_{d,b,t} = \log(1 + NumTrades_{d,b,t})$ measures trade intensity.
- Ask and bid quote indicators distinguish one-sided and two-sided quoting.

Interpretation: The extensive margin asks whether the dealer participates at all. The intensive margin asks whether more frequent quoting is associated with more trading.

3.5 Execution quality

Client execution quality is measured relative to intraday benchmarks:

$$ClientExecutionQuality_{d,b,t,i} = \begin{cases} BenchmarkPrice_{b,t,i} - Price_{d,b,t,i}, & \text{client buy,} \\ Price_{d,b,t,i} - BenchmarkPrice_{b,t,i}, & \text{client sell.} \end{cases} \quad (1)$$

- Higher values mean better client execution.
- Benchmarks are based on market prices around the trade or dealer quote benchmarks.
- The measure is expressed in dollars per \$100 par value, or equivalently in basis-point-like units for interpretation.

Interpretation: The measure converts transaction prices into a common measure of how favorable the trade was for the client.

3.6 Quote quality

The paper defines quote quality relative to benchmark quotes:

$$BidQuoteQuality_{d,b,t} = BidQuote_{d,b,t} - BenchBidQuote_{b,t}, \quad (2)$$

$$AskQuoteQuality_{d,b,t} = BenchAskQuote_{b,t} - AskQuote_{d,b,t}. \quad (3)$$

- A positive quote quality value means the dealer posted a more aggressive quote than the benchmark.
- Bid and ask quality are averaged when both sides are available.
- Quote quality is set to zero when the dealer does not quote or when the benchmark is not defined.

Interpretation: Quote quality measures whether a dealer is advertising a better trading price than competing dealers.

4 Empirical specifications and interpretation

4.1 Determinants of quoting

The baseline quoting regression is:

$$HasQuote_{d,b,t} = \alpha_d + \alpha_t + \beta_1 DealerType_{d,b} + \beta_2 InventoryManagement_{d,b,t-1-h,t-1} + \beta_3 BondType_{b,t-1} + \beta_4 InformationEvent_{d,b,t-1} \quad (4)$$

- *DealerType* captures lead underwriter and underwriting-role indicators.
- *InventoryManagement* captures dealer and connected-dealer inventory changes.
- *BondType* includes rating, maturity, bond age, institutional investor count, issuance size, and bond type.
- *InformationEvent* captures rating changes around the quote date.

Interpretation: The regression asks why dealers choose to quote. The paper finds evidence consistent with advertising and inventory management, but little evidence that quote supply is driven by private information about rating changes.

4.2 Dealer-level quote-trade relation

The dealer-level saturated fixed-effect regression is:

$$HasTrade_{d,b,t} = \alpha_{d \times b} + \alpha_{d \times t} + \alpha_{b \times t} + \beta Q_{d,b,t} + \gamma HasTrade_{d,b,t-1} + \varepsilon_{d,b,t}, \quad (5)$$

where $Q_{d,b,t}$ is either $HasQuote_{d,b,t}$ or $LogNumQuotes_{d,b,t}$.

Interpretation: Dealer-bond fixed effects absorb stable dealer specialization in a bond. Dealer-day fixed effects absorb dealer-wide inventory or trading shocks. Bond-day fixed effects absorb bond-specific market conditions. The coefficient β therefore compares dealers in the same bond-day and asks whether the quoting dealer is more likely to trade.

4.3 Bond-level quote supply and trading

At the bond-day level, the paper estimates:

$$Y_{b,t} = \alpha_b + \alpha_t + \beta QuoteSupply_{b,t}^{ext/int} + \varepsilon_{b,t}, \quad (6)$$

where $Y_{b,t}$ is $CliQShare_{b,t}$, $\log(1 + CliQDealerTrades_{b,t})$, or $\log(1 + TotalTrades_{b,t})$.

- $QuoteSupply^{ext}$ measures the fraction or amount of dealers quoting a bond.
- $QuoteSupply^{int}$ measures the intensity of quoting.
- The authors also construct shift-share instruments using dealer-level changes in quote supply interacted with lagged dealer-bond weights.

Interpretation: The bond-level analysis asks whether quoting expands market activity, not just whether individual dealers steal trades from one another.

4.4 Quote presence and client execution quality

At the trade level:

$$ClientExecutionQuality_{d,b,t,i} = \alpha_{d \times b} + \alpha_{d \times t} + \alpha_{b \times t} + \beta Q_{d,b,t} + \varepsilon_{d,b,t,i}. \quad (7)$$

Interpretation: A positive β means that clients obtain better prices when trading with a dealer that has posted quotes in the bond on that day.

4.5 Quote quality and execution quality

The quote-quality regression is:

$$ClientExecutionQuality_{d,b,t,i} = \alpha_{d \times b} + \alpha_{d \times t} + \alpha_{b \times t} + \beta QuoteQuality_{d,b,t} + \delta HasQuote_{d,b,t} + \varepsilon_{d,b,t,i}. \quad (8)$$

The paper further separates positive and negative quote quality:

$$QuoteQuality_{d,b,t} = QuoteQuality_{d,b,t}^+ \mathbf{1}\{QuoteQuality_{d,b,t} \geq 0\} + QuoteQuality_{d,b,t}^- \mathbf{1}\{QuoteQuality_{d,b,t} < 0\}. \quad (9)$$

Interpretation: This split tests whether good quotes are firm and whether bad quotes are improved in bargaining. The paper finds that good quotes pass through strongly to execution quality, while bad quotes have weaker effects because dealers improve their quoted prices during negotiation.

4.6 Relationship substitution

The insurer-dealer matching regression is:

$$Pr(NRDTrade_{c,d,b,t,i}) = F(\alpha + \beta_{NRD}NRDQuote_{c,b,t} + \beta_{RD}RDQuote_{c,b,t} + \beta_{NRD \times RD}NRDQuote_{c,b,t} \times RDQuote_{c,b,t} + \gamma) \quad (10)$$

Interpretation: The dependent variable equals one when an insurer trades with a nonrelationship CliQ dealer rather than a relationship CliQ dealer. A positive β_{NRD} means nonrelationship dealers attract clients by quoting.

5 Identification and empirical design

5.1 Within dealer-bond-day logic

The most demanding specifications use dealer-bond, dealer-day, and bond-day fixed effects. These fixed effects compare dealers trading the same bond on the same day while absorbing stable dealer specialization and daily dealer-wide shocks.

Interpretation: This structure substantially reduces concern that better dealers or more liquid bonds mechanically have both more quotes and more trades.

5.2 Inventory and underwriting motives

- Inventory changes help identify whether dealers quote to manage positions.
- Underwriting role identifies dealers with specialized knowledge and bond-specific liquidity provision responsibilities.
- Rating events help test whether quote supply is driven by private credit information.

Interpretation: The paper finds that quoting is connected to trading and inventory motives rather than primarily to private rating-change information.

5.3 Shift-share approach at the bond level

- Bond-level quote supply may be endogenous to bond-level trading interest.
- The paper uses dealer-level changes in quoting across all bonds as shifts.
- It weights those shifts by the dealer’s lagged trading importance in the bond.

Interpretation: This approach separates market-wide dealer quoting shocks from bond-specific trading demand shocks.

5.4 Insurance-company relationship tests

- Relationship and nonrelationship dealers are defined using pre-sample insurer-dealer trading patterns.
- The paper then asks whether quotes by nonrelationship dealers affect the probability that insurers choose them.
- The design isolates quote-driven switching away from established relationships.

Interpretation: This is direct evidence that quotes can substitute for relationship-based search and matching.

6 Tables and figures

6.1 Table I: Descriptive statistics and Figure 1: Intraday quote and trade distribution

Read:

- Table I shows that the median quote message contains seven bonds and that 27 dealers quote per day on average.
- CliQ dealers trade heavily, with 34 dealers trading per day and roughly 1.64 CliQ dealers trading a given bond per day.
- Panel C shows that at the dealer-bond-day level, a dealer quotes 9% of the time and trades 5% of the time.
- Figure 1 shows that quotes arrive earlier in the day than trades: quoting activity peaks in the morning, while trades peak in the afternoon.

Economic message: Dealer quotes are a meaningful pre-trade information flow. Their timing suggests that quote dissemination precedes trading and can help organize later order flow.

Table I
Descriptive Statistics

The table provides descriptive statistics about individual dealer quotes and trades—number per day, frequency, quality, and intraday timing.

Variable	<i>N</i>	Mean	<i>SD</i>	5%	25%	50%	75%	95%
Panel A: Quotes by CliQ dealers								
No. dealers quoting per day (on days with quotes)	146	27.03	2.75	22	26	27	29	31
No. dealers quoting per bond-day (conditional on bond quote)	631,435	1.93	1.23	1	1	2	2	4
No. bonds quoted per day (on days with quotes)	146	4,324.90	950.95	2,499	3,781	4,711	5,034	5,387
No. bonds quoted per dealer-day (conditional on dealer quote)	3,946	309.62	412.08	6	41	153	413	1,171
No. bonds quoted per message	179,531	13.71	25.57	1	3	7	16	42
No. messages per dealer-day (conditional on dealer message)	3,946	45.50	81.58	1	5	13.50	40	236
Quoted bid-ask spread if available (winsorized at 0.5% and 99.5%)	1,931,288	0.74	0.65	0.09	0.26	0.56	1.00	2.00
Quote time-of-day (hour)	2,462,193	10.64	3.10	7.25	8.32	9.85	12.57	16.42
Panel B: Trades by CliQ dealers								
No. dealers trading per day (on days with trades)	146	34.13	1.09	32	34	34	35	35
No. dealers trading per bond-day (conditional on bond trade)	413,132	1.64	1.02	1	1	1	2	4
No. bonds traded per day (on days with trades)	146	2,829.67	357.38	2,129	2,725	2,930	3,040	3,182
No. bonds traded per dealer-day (conditional on dealer trade)	4,983	135.73	201.35	2	15	53	164	695
Trade time-of-day (hour)	1,202,957	12.95	2.39	8.97	11.10	13.20	15.00	16.35
Panel C: Quotes and trades at dealer-bond-day level, balanced panel								
<i>HasQuote_{d,b,t}</i>	14,031,768	0.09	0.28	0.00	0.00	0.00	0.00	1.00
<i>HasTrade_{d,b,t}</i>	14,031,768	0.05	0.21	0.00	0.00	0.00	0.00	0.00
<i>LogNumQuotes_{d,b,t}</i>	14,031,768	0.08	0.30	0.00	0.00	0.00	0.00	0.69
<i>LogNumTrades_{d,b,t}</i>	14,031,768	0.04	0.21	0.00	0.00	0.00	0.00	0.00

(a) Table I: Descriptive statistics

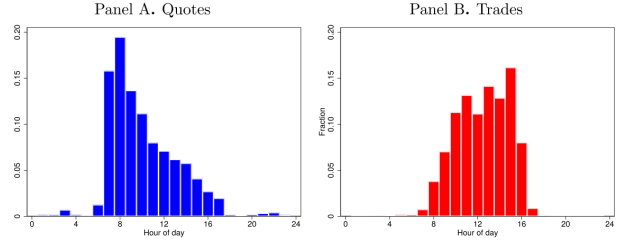


Figure 1. Distribution of quotes and trades during the day. The figure documents the distribution of trades and quotes during the day. We pool all dealer bond days and plot the fraction f of the quotes (Panel A) and trades (Panel B) that arrive during a given hour. (Color figure can be

(b) Figure 1: Intraday quote and trade distribution

6.2 Table II: Determinants of quoting, part 1 and Table II: Determinants of quoting, continued

Read:

- Table II shows that lead underwriters are more likely to quote.
- Own inventory changes predict quote supply, while connected dealer inventory is much less important.
- Higher-rated bonds, older bonds, and larger issuance bonds are more likely to be quoted.
- Rating changes are not strong predictors of quote supply.

Economic message: Quotes appear to reflect dealer specialization, inventory management, and trading intent rather than primarily hidden information about credit events.

Table II
Determinants of Quoting

The table reports results on the determinants of dealers' decisions to quote at the ask or bid, ask only, and bid only, where $h = 20$ days. Estimates are obtained from panel regressions with dealer and day fixed effects. All coefficients are scaled by 100. Standard errors are double-clustered at the dealer and bond levels. Significance levels: *** 1%, ** 5%, * 10%.

Dependent Variable:	$HasQuote_{d,h,t}$	$HasAskQuote_{d,h,t}$		$HasBidQuote_{d,h,t}$	
		All (1)	One-Sided Ask (3)	All (4)	One-Sided Bid (5)
<i>Dealertype_{d,t}</i>					
Lead underwriter	2.688** (1.081)	2.618** (1.023)	0.069 (0.062)	2.631** (1.079)	0.082 (0.106)
Underwriter	1.451 (1.009)	1.458 (1.069)	-0.068 (0.073)	1.533 (1.008)	0.007 (0.089)
<i>Inventorymanagement_{d,h,t-1,h,t-1}</i>					
Signed inventory change	0.179** (0.077)	0.220** (0.092)	0.126*** (0.042)	0.050 (0.065)	-0.044 (0.027)
Connected inventory change	-0.185 (0.260)	-0.190 (0.249)	-0.020* (0.016)	-0.157 (0.260)	0.004 (0.014)
No inventory change	-3.186*** (1.105)	-2.654*** (0.944)	-0.951** (0.355)	-2.230** (0.857)	-0.527** (0.252)
<i>Bondtype_{d,t-1}</i>					
Rating (1 = AAA, 2 = AA+, ...)	-0.617*** (0.163)	-0.537*** (0.150)	-0.061 (0.038)	-0.546*** (0.154)	-0.069*** (0.022)
Log years to mature	1.249* (0.671)	1.392* (0.686)	-0.123 (0.119)	1.424** (0.660)	-0.092 (0.066)
Log age of bond	0.807*** (0.204)	0.744*** (0.196)	0.049 (0.037)	0.753*** (0.195)	0.058* (0.030)
Log no. institutional investors	-0.148 (0.572)	-0.130 (0.525)	-0.155 (0.145)	-0.010 (0.547)	0.025 (0.080)
Log size of issuance	0.812** (0.318)	0.823** (0.305)	0.251* (0.131)	0.562* (0.285)	-0.010 (0.037)

(Continued)

(a) Table II: Determinants of quoting, part 1

Table II—Continued

Dependent Variable:	$HasQuote_{d,h,t}$	$HasAskQuote_{d,h,t}$		$HasBidQuote_{d,h,t}$	
		All (1)	One-Sided Ask (3)	All (4)	One-Sided Bid (5)
MTN	0.364 (0.684)	0.242 (0.621)	0.180 (0.166)	0.189 (0.666)	0.127 (0.124)
Rule 144a bond	0.976 (0.640)	0.898 (0.579)	-0.114 (0.136)	1.071* (0.585)	0.059 (0.114)
Private placement	0.584 (1.282)	0.818 (1.315)	-0.230 (0.186)	0.832 (1.272)	-0.216** (0.104)
<i>Informationevent_{d,t,t+h}</i>					
Rating change in next 5 days	-0.733 (0.532)	-0.737 (0.515)	-0.059 (0.078)	-0.676 (0.524)	0.002 (0.068)
Fixed effects	Dealer, Date	Dealer, Date	Dealer, Date	Dealer, Date	Dealer, Date
R ²	0.129	0.130	0.029	0.131	0.012
N	14,031,768	14,031,768	14,031,768	14,031,768	14,031,768

(b) Table II: Determinants of quoting, continued

6.3 Table III: Individual quoting and trading activity and Table IV: Dealer quote activity and order flow

Read:

- Table III shows that trading probability is 6.7% when a dealer quotes versus 4.6% when the dealer does not quote.
- The quote-trade relation is stronger for institution-sized trades.
- Table IV estimates saturated fixed-effect regressions and finds that quote presence raises dealer trade probability by about 4.8 percentage points.
- Quoting intensity also increases the dealer's number of trades.

Economic message: Quotes are strongly associated with subsequent order flow even after controlling for dealer-bond, dealer-day, and bond-day factors.

Table III
Individual Quoting and Trading Activity

The table reports results on the relationship between individual dealers' quoting and trading. We aggregate quotes and trades by each CliQ dealer at the bond-day level. Standard errors are triple-clustered at the dealer, bond, and day levels. Significance levels: *** 1%, ** 5%, * 10%.

Panel A: Probability of trade by dealer d in bond b on day t					
Variable	N	All dealer-bond-days	$HasQuote_{d,b,t} = 0$	$HasQuote_{d,b,t} = 1$	Δ (1-0)
$HasTrade_{d,b,t}$	14,031,768	4.8%	4.6%	6.7%	2.1pps*
Trade size $\geq \$100K$		2.4%	2.2%	4.6%	2.4pps***
Trade size $< \$100K$		2.8%	2.8%	2.8%	0.0pps
Client trade		2.5%	2.4%	3.6%	1.2pps*
Interdealer trade		3.3%	3.3%	4.2%	0.9pps
$HasTrade_{d,b,t}$, split by:					
IG rated	11,286,354	4.5%	4.3%	6.5%	2.1pps*
HY rated	2,745,414	5.9%	5.8%	10.8%	4.9pps*
Panel B: Probability of afternoon trade by dealer d in bond b on day t					
Variable	N	All Dealer-bond-days	$HasQuote_{d,b,t}^{AM} = 0$	$HasQuote_{d,b,t}^{AM} = 1$	Δ (1-0)
$HasTrade_{d,b,t}^{PM}$	14,031,768	3.3%	3.2%	4.6%	1.4pps*
Trade size $\geq \$100K$		1.6%	1.5%	3.0%	1.5pps***
Trade size $< \$100K$		1.9%	1.9%	2.0%	0.1pps
$HasTrade_{d,b,t}^{PM}$, split by:					
IG rated	11,286,354	3.1%	3.0%	4.5%	1.5pps*
HY rated	2,745,414	4.0%	4.0%	6.9%	2.9pps

(a) Table III: Individual quoting and trading activity

Table IV
Dealer's Quoting Activity and Order Flow

The table reports results on the determinants of dealer trades. In Panel A, the dependent variable equals one if dealer d trades bond b on day t , and zero otherwise. In Panel B, the dependent variable equals the natural logarithm of the number of trades plus one by dealer d in bond b on day t . Estimates are obtained from panel regressions with saturated dealer-bond, dealer-day, and bond-day fixed effects. The number of observations before (after) dropping singleton observations is 14,031,768 (13,935,660). Standard errors are triple-clustered at the dealer, bond, and day levels. Significance levels: *** 1%, ** 5%, * 10%.

Panel A: Dealer's trading activity, extensive margin				
Dependent Variable	$HasTrade_{d,b,t}$	$HasTrade_{d,b,t}$	$HasTrade_{d,b,t}$	$HasTrade_{d,b,t}$
	(1)	(2)	(3)	(4)
$HasQuote_{d,b,t}$	0.048*** (0.008)		0.043*** (0.007)	
$LogNumQuotes_{d,b,t}$		0.060*** (0.008)		0.053*** (0.007)
$HasTrade_{d,b,t-1}$			0.146*** (0.008)	0.145*** (0.007)
Fixed effects	Saturated	Saturated	Saturated	Saturated
R^2	0.256	0.257	0.272	0.273
N	14,031,768	14,031,768	13,935,660	13,935,660
Panel B: Dealer's trading activity, intensive margin				
Dependent variable:	$LogNumTrades_{d,b,t}$	$LogNumTrades_{d,b,t}$	$LogNumTrades_{d,b,t}$	$LogNumTrades_{d,b,t}$
	(1)	(2)	(3)	(4)
$HasQuote_{d,b,t}$	0.044*** (0.008)		0.038*** (0.007)	
$LogNumQuotes_{d,b,t}$		0.055*** (0.008)		0.048*** (0.007)
$LogNumTrades_{d,b,t-1}$			0.174*** (0.011)	0.172*** (0.011)
Fixed effects	Saturated	Saturated	Saturated	Saturated
R^2	0.277	0.278	0.299	0.300
N	14,031,768	14,031,768	13,935,660	13,935,660

(b) Table IV: Dealer quote activity and order flow

6.4 Table V: Bond-level quote supply, part 1 and Table V: Bond-level quote supply, continued

Read:

- Table V studies bond-day quote supply rather than individual dealer quotes.
- The OLS and IV results show that more quote supply increases CliQ dealers' market share, the number of CliQ dealer trades, and total bond-level trades.
- The second page of Table V shows that total trades rise with quote supply, not just CliQ trades.

Economic message: Quote competition is not just reallocation across dealers. More active quoting is associated with more total trading in the bond.

Table V
Bond-Level Quote Supply and Trading Activity

The table reports results on the relationship between CliQ dealers' aggregate quote supply and trading activity by CliQ dealers and non-CliQ dealers. We focus on institutional-sized trades between dealers and clients, and in falsification tests, we focus on retail-sized trades. We split trades by CliQ dealers versus non-CliQ dealers and collapse all trades at the bond-day level. In columns (1) and (2), we use $QuoteSupply_{b,t}^{ext/int}$ as the explanatory variable in the OLS specification (4). In columns (3) and (4), we use the dealer's share of institutional-sized trades from (B2) as our measure of dealers' importance and the dealer's aggregate daily change in quotes from (B3) as quote supply in bond b on day t to construct our instrument and report the IV second-stage coefficient estimates on the predicted $QuoteSupply_{b,t}^{ext/int}$. Estimates are obtained from panel regressions with bond and day fixed effects. Standard errors are double-clustered at the bond and day levels. Significance levels: *** 1%, ** 5%, * 10%.

Panel A: Quotes and trades at bond-day level, balanced panel								
Variable	N	Mean	SD	5%	25%	50%	75%	95%
$QuoteSupply_{b,t}^{ext}$	1,406,564	0.07	0.19	0.00	0.00	0.00	0.00	0.50
$QuoteSupply_{b,t}^{int}$	1,406,564	0.19	0.74	0.00	0.00	0.00	0.00	1.04
$CliQShare_{b,t}$	530,576	0.20	0.34	0.00	0.00	0.00	0.33	1.00
$\log(CliQDealerTrades_{b,t} + 1)$	1,406,564	0.11	0.32	0.00	0.00	0.00	0.00	0.69
$\log(TotalTrades_{b,t} + 1)$	1,406,564	0.42	0.62	0.00	0.00	0.00	0.69	1.61

Panel B: Impact of dealer quotes on CliQ dealer trading and total trading				
	OLS		2SLS	
	(1)	(2)	(3)	(4)
Dependent variable: $CliQShare_{b,t}$				
$QuoteSupply_{b,t}^{ext}$	0.151*** (0.004)		0.107** (0.050)	
$QuoteSupply_{b,t}^{int}$		0.031*** (0.001)		0.015* (0.008)
Fixed effects	Bond, Date	Bond, Date	Bond, Date	Bond, Date
R^2	0.099	0.097	0.007	0.004
N	530,576	530,576	530,576	530,576

Dependent variable: $\log(CliQDealerTrades_{b,t} + 1)$				
$QuoteSupply_{b,t}^{ext}$	0.150*** (0.004)		0.107* (0.062)	
$QuoteSupply_{b,t}^{int}$		0.042*** (0.002)		0.029** (0.013)
Fixed effects	Bond, Date	Bond, Date	Bond, Date	Bond, Date
R^2	0.180	0.181	0.007	0.008
N	1,406,564	1,406,564	1,406,564	1,406,564

(Continued)

(a) Table V: Bond-level quote supply, part 1

Table V—Continued

Dependent variable: $\log(TotalTrades_{b,t} + 1)$				
$QuoteSupply_{b,t}^{ext}$	0.201*** (0.010)		0.244* (0.145)	
$QuoteSupply_{b,t}^{int}$		0.051*** (0.003)		0.075** (0.030)
Fixed effects	Bond, Date	Bond, Date	Bond, Date	Bond, Date
R^2	0.425	0.425	0.005	0.004
N	1,406,564	1,406,564	1,406,564	1,406,564

(b) Table V: Bond-level quote supply, continued

6.5 Table VI: Execution quality and dealer quotes and Table VII: Execution quality and quote quality

Read:

- Table VI shows that clients obtain about 7 to 9 basis points better execution when trading with a dealer that quotes.
- Bond-level quote supply also improves bond-level execution quality.
- Table VII shows strong pass-through from quote quality to execution quality: clients trading with dealers posting better quotes receive better prices.
- The pass-through is especially strong for positive quote quality, meaning good quotes are informative and relatively firm.

Economic message: Quotes affect prices, not just matching. Better displayed terms translate into lower client trading costs.

Table VI
Client Execution Quality and Dealer Quotes

Panel A reports descriptive statistics on client execution quality. Panel B documents the relationship between CliQ dealers' quotes and trade-level client execution quality. Estimates are obtained from panel regressions with saturated dealer-bond, dealer-day, and bond-day fixed effects. Standard errors are triple-clustered at the dealer, bond, and day levels. Panel C documents the relationship between CliQ dealers' quote supply and bond-level client execution quality. We split trades by CliQ dealers versus non-CliQ dealers and collapse all trades at the bond-day level. Estimates are obtained from 2SLS and OLS regressions with bond and day fixed effects. Standard errors are double-clustered at the bond and day levels. Significance levels: *** 1%, ** 5%, * 10%.

Panel A: Descriptive statistics on $ClientExecutionQuality_{d,b,t,i}$								
Variable	N	Mean	SD	5%	25%	50%	75%	95%
$ClientExecutionQuality_{d,b,t,i}$ (\$):								
BAML benchmark	417,512	-0.37	1.75	-2.87	-0.64	-0.15	0.13	1.43
Interdealer benchmark	452,235	-0.29	1.18	-2.00	-0.46	-0.08	0.03	0.81
$ClientExecutionQuality_{d,b,t,i}$ split by:								
Trade size $\geq \$100K$	233,106	-0.24	1.79	-2.72	-0.53	-0.12	0.19	1.81
Trade size $< \$100K$	184,406	-0.54	1.67	-3.02	-0.81	-0.19	0.07	0.85
IG rated	303,876	-0.36	1.69	-2.71	-0.55	-0.13	0.12	1.16
HY rated	113,636	-0.40	1.88	-3.18	-0.88	-0.22	0.16	2.09

Panel B: Impact of dealer quotes on trade-level client execution quality			
Dependent variable: $ClientExecutionQuality_{d,b,t,i}$			
	(1)	(2)	(3)
$HasQuote_{d,b,t}$	0.069** (0.029)		0.089* (0.045)
$LogNumQuotes_{d,b,t}$		0.035 (0.024)	-0.020 (0.036)
Fixed effects	Saturated	Saturated	Saturated
R^2	0.489	0.489	0.489
N	417,512	417,512	417,512

Panel C: Impact of dealer quotes on bond-level client execution quality				
Dependent variable: $BondExecutionQuality_{b,t}$				
	OLS		2SLS	
	(1)	(2)	(3)	(4)
$QuoteSupply_{b,t}^{ext}$	0.066*** (0.024)		0.794** (0.386)	
$QuoteSupply_{b,t}^{int}$		0.008** (0.003)		0.101** (0.044)
Fixed effects	Bond, Date	Bond, Date	Bond, Date	Bond, Date
R^2	0.103	0.103	-0.009	-0.004
N	154,528	154,528	154,528	154,528

(a) Table VI: Execution quality and dealer quotes

Table VII
Client Execution Quality and Quote Quality

The table reports results on the relationship between client execution quality and quote quality. Panel A is restricted to the sample in which client execution quality is not missing and there is at least a quote. Estimates are obtained from panel regressions with saturated dealer-bond, dealer-day, and bond-day fixed effects. Standard errors are triple-clustered at the dealer, bond, and day levels. Significance levels: *** 1%, ** 5%, * 10%.

Panel A: Descriptive statistics on $QuoteQuality_{d,b,t}$								
Variable	N	Mean	SD	5%	25%	50%	75%	95%
$QuoteQuality_{d,b,t}$ (\$):								
BenchQuote = Average quote	49,727	0.01	0.15	-0.21	0.00	0.00	0.02	0.29
BenchQuote = Best quote	49,727	-0.14	0.28	-0.81	-0.14	0.00	0.00	0.00

Panel B: Impact of dealer's quote quality on client execution quality				
Dependent variable: $ClientExecutionQuality_{d,b,t,i}$				
Sample:	All days (1)	$HasQuote_{d,b,t}=1$ (2)	All days (3)	$HasQuote_{d,b,t}=1$ (4)
$QuoteQuality_{d,b,t}$	0.838*** (0.180)	1.102*** (0.216)		
$QuoteQuality_{d,b,t}^+$			1.202*** (0.199)	1.389*** (0.166)
$QuoteQuality_{d,b,t}^-$			0.385 (0.349)	0.730 (0.627)
$HasQuote_{d,b,t}$	0.066** (0.030)		0.037 (0.030)	
Fixed effects	Saturated	Saturated	Saturated	Saturated
R^2	0.489	0.418	0.489	0.418
N	417,512	49,727	417,512	49,727

(b) Table VII: Execution quality and quote quality

6.6 Table VIII: Trade-throughs and price improvement and Table IX: Relationship versus nonrelationship dealers

Read:

- Table VIII shows that trade-throughs are frequent: trades often occur at prices worse than the best available quote.
- Price improvement is larger when the dealer's posted quote is worse, indicating negotiation and price matching.
- Table IX shows that relationship dealers are more common trading counterparties, but nonrelationship dealers have a higher quote rate conditional on being chosen.
- Nonrelationship dealers handle 23% of trades when they quote versus 11% when they do not quote.

Economic message: OTC quote competition remains imperfect because clients and dealers negotiate after quotes are observed. However, quotes still reduce the dominance of existing relationships.

Table VIII
Trade-Throughs and Price Improvement over Quote

The table reports results on trade-throughs and the price improvement over the dealer's quote defined in (11) at the dealer-bond-day level. Panel A reports the fraction of trades that occur at a price worse than the best dealer quote that is less than one hour old. Fractions are calculated by small ($< \$100K$) and large ($\geq \$100K$) trade sizes for the best-quoting dealer, CliQ dealers who are not quoting or quoting at an inferior price, and non-CliQ dealers. In Panels B and C, we use the last quote provided by the dealer as the relevant dealer quote. Trades not preceded by a quote from the same dealer in the same bond on the same day are eliminated since we cannot compute price improvement. For Panel C, we further restrict the sample to cases in which the trading dealer has at least a quote and at least one other dealer provides a quote.

Panel A: Trade occurs at price worse than best quote								
Sample:	Best Quote Dealer	CliQ Dealer, Not Best Quote	Non-CliQ Dealer					
Trade – through $_{d,b,t,i}$ (%/100)	0.17	0.31	0.33					
Trade size $\geq \$100K$	0.16	0.25	0.26					
Trade size $< \$100K$	0.23	0.40	0.39					

Panel B: Descriptive statistics on $PriceImprovement_{d,b,t,i}$								
Variable	N	Mean	SD	5%	25%	50%	75%	95%
$PriceImprovement_{d,b,t,i}$ (\$)	40,844	0.07	0.75	−0.75	−0.04	0.03	0.19	0.89
Trade size $\geq \$100K$	31,567	0.12	0.71	−0.45	−0.02	0.04	0.21	0.95
Trade size $< \$100K$	9,277	−0.11	0.83	−1.63	−0.12	0.00	0.12	0.74
$PriceImprovement_{d,b,t,i}$ split by:								
IG rated	37,516	0.07	0.74	−0.72	−0.04	0.03	0.18	0.88
HY rated	3,328	0.08	0.75	−1.09	−0.03	0.05	0.25	1.03

Panel C: $PriceImprovement_{d,b,t,i}$ split by quote quality								
Quartile of $QuoteQuality_{d,b,t}$	N	Mean	SD	5%	25%	50%	75%	95%
Low	8,230	0.18	0.87	−0.81	−0.02	0.09	0.34	1.27
2	6,277	0.04	0.60	−0.44	−0.03	0.02	0.11	0.52
3	8,012	0.05	0.36	−0.20	−0.02	0.02	0.10	0.43
High	8,654	−0.01	0.83	−1.29	−0.12	0.02	0.21	0.86

(a) Table VIII: Trade-throughs and price improvement

Table IX
Quoting and Trading with (Non)Relationship Dealers

Panel A reports ex ante quote rates across relationship and nonrelationship CliQ dealers. For each insurer trade, all possible counterparties (relationship and nonrelationship dealers) are considered. Panel B reports quote rates depending on whether insurers choose to trade with relationship or nonrelationship CliQ dealers. The sample consists of all trades between insurers and CliQ dealers. Relationship dealers are defined as those in the top half of counterparties of a given insurer from January 1, 2018, to September 30, 2019.

Panel A: Ex ante quote rates (across all possible dealer counterparties for each insurer trade)				
Variable	Total	Quote Rate		
Relationship dealers ($RDQuote_{c,b,t}$)	1,375,309	4.4%		
Nonrelationship dealers ($NRDQuote_{c,b,t}$)	1,054,520	3.6%		

Panel B: Trades and quote rates (across all insurer trades)				
Sample:	Total	Quoted	Not Quoted	Quote Rate
All trades	107,523	11,797	95,726	11.0%
Relationship dealers ($RDTrade_{c,d,b,t,i}$)	94,448	9,041	85,407	9.6%
Nonrelationship dealers ($NRDTrade_{c,d,b,t,i}$)	13,075	2,756	10,319	21.1%

(b) Table IX: Relationship versus nonrelationship dealers

6.7 Table X: Quoting and client-dealer matching and Appendix A: data filters

Read:

- Table X reports logit models of trading with a nonrelationship dealer.
- Nonrelationship dealer quotes strongly increase the probability that the insurer trades with a nonrelationship dealer.
- Relationship dealer quotes have a negative or insignificant effect in the same regressions.
- Appendix A documents how the raw quote sample is filtered from 8.5 million quotes to the analysis sample.

Economic message: Quotes help dealers win new clients and substitute partly for relationships. The data filters show that the final sample is carefully constructed from raw runs and matched to usable price information.

Table X
Quoting and Client-Dealer Matching

The table reports logit regression results for the probability of trading with a nonrelationship CliQ dealer (1) as opposed to a relationship CliQ dealer (0). The sample consists of all trades between insurers and CliQ dealers. Relationship dealers are defined as those in the top half of counterparties of the same insurer from January 1, 2018 to September 30, 2019. Standard errors are clustered at the insurer level. Significance levels: *** 1%, ** 5%, * 10%.

Dependent Variable:	$Pr(NRDTrade_{c,d,b,t,i})$		$Pr(FirstNRDTrade_{c,d,b,t,i})$	
	(1)	(2)	(3)	(4)
$NRDQuote_{c,b,t}$	4.417*** (0.623)	5.126*** (0.766)	3.824*** (0.506)	4.129*** (0.612)
$RDQuote_{c,b,t}$	−1.517*** (0.589)	−0.789 (0.498)	−0.815 (0.761)	−0.488 (0.682)
$NRDQuote_{c,b,t} \times RDQuote_{c,b,t}$		−9.083*** (2.889)		−4.478* (2.687)
Controls	Yes	Yes	Yes	Yes
Pseudo R^2	0.053	0.054	0.067	0.067
N	107,523	107,523	107,523	107,523

(a) Table X: Quoting and client-dealer matching

Appendix A: Data Filters and Yield Spread-Price Conversion

The following table presents the steps to filter the CliQ data on dealer quotes. Quotes on the bid side and the ask side are treated as separate quotes.

Step	No. Quotes	No. Bonds
Original (Oct 1, 2019–May 1, 2020)	8,546,249	24,441
In FISD	6,526,786	13,477
Keep bond type "CDEB", "CZ", "CPIK", "CCOV", "CLOC", "UCID", "CMTN", "CMTZ", "EBON", "EMTN"	6,045,106	12,102
Not exchangeable, preferred or convertible	6,042,870	12,091
Drop government type or foreign	4,584,077	9,678
Drop if missing offering date, maturity, offering amt or offering amt $< \$100K$	4,583,478	9,670
Drop floating rate	4,312,326	8,764
Drop maturity $< 1/1/2021$ or maturity $> 12/31/2070$		
or payment-in-kind type coupon	4,229,643	8,113
Drop if price = spread	4,187,757	8,101
Drop bond trading holiday	4,183,125	8,101
Drop non-unique observation at the timestamp, firm, CUSIP, and quote side	4,170,931	8,101
Keep semi-annual interest payment or zero coupon	4,160,948	8,077

(b) Appendix A: data filters

7 Short summary

- Dealer runs are an important form of pre-trade transparency in corporate bond markets.
- Quotes arrive earlier than trades and often advertise dealer trading interest.
- Dealers quote more when they have inventory motives or underwriting specialization.
- Quoting increases the probability and intensity of dealer-level trading.
- Bond-level quote supply increases total bond-level trading.
- Clients receive better execution when they trade with dealers that quote and with dealers offering better quote quality.
- The best quote does not always win the trade, and trade-throughs are common.
- Dealers with worse quotes often improve prices in negotiation.
- Quotes partly substitute for client-dealer relationships by allowing nonrelationship dealers to attract order flow.
- The paper shows that indicative, nonbinding quotes can still improve OTC market efficiency, even though quote competition remains imperfect.

8 Conclusion

8.1 Core conclusion

Hendershott, Li, Livdan, Schürhoff, and Venkataraman show that dealer quotes in corporate bonds matter. Even though these quotes are indicative and nonbinding, they increase trading, lower client trading costs, and help nonrelationship dealers compete for order flow.

8.2 Economic intuition

In an opaque OTC market, clients face search and bargaining frictions. Dealer quotes reduce those frictions by advertising a dealer’s willingness to trade and by giving clients outside options in negotiation. However, because quotes are not binding and are not universally observed, quote competition is imperfect. Clients can use quotes to bargain, dealers can improve bad quotes, and the best quote does not always win.

8.3 Incremental contribution takeaway

The paper adds direct pre-trade evidence to the corporate bond market literature. Instead of inferring liquidity from trades alone, it shows how quotes shape order flow, execution quality, and relationship formation. Its main incremental contribution is to demonstrate that partial pre-trade transparency can improve OTC market functioning while still leaving substantial room for search frictions and relationship-based bargaining.